Chapter 1

Laser Radiation and its properties

LASER is an acronym for **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation. **A laser is an optical device that transforms energy from external sources into a coherent, highly monochromatic, intense, and highly directional beam of electromagnetic radiation through the quantum process of stimulated emission**.

1.1 Historical background of laser

The fundamental ideas of **stimulated emission amplification** found their first realization in the microwave region in the 1950s in the form of the **maser** invention. The **MASER** (**M**icrowave **A**mplification by **S**timulated **E**mission of **R**adiation) was invented by Charles H. Townes from the USA, Alexander M. Prokhorov and Nikolai G. Basov from the USSR.

The development of maser technology in the optical spectrum was limited by the physical constraints postulated by Arthur L. Schawlow and Charles H. Townes in 1958 in their formulation of optical formulation.

In 1960, Dr. Theodore H. Maiman built the first optical laser that was actually functional by pumping a synthetic solid-state ruby crystal with a helical flashlamp.

Modern advances (1961-present): Active media, such as Nd:YAG, semiconductor laser diodes (e.g., GaAs), gas mixtures (e.g., He-Ne, liquid dyes, and optical fibers were rapidly developed. Currently, research is aimed at the generation of ultra-short pulses (up to femtosecond regimes down to(4.5 fs or $4.5*10^{-15}$ seconds) and high continuous power output of more than 10 kW.

Uses of laser technology are: Optical communication devices, lasers for material processing (welding, cutting), non-destructive testing, optical data storage, medical surgery, high resolution spectroscopy, positioning control, and holography.

1.2 Electromagnetic waves and its property

Electromagnetic radiation is one of the many ways that energy travels through space. The heat from a burning fire, the light from the sun, the X-rays used by your doctor, as well as the energy used to cook food in a microwave are all forms of electromagnetic radiation. While these forms of energy might seem quite different from one another, they are related in that they all exhibit wavelike properties.

Waves are simply disturbances in a medium or a field, resulting in a vibration or oscillation transferring energy from one point to another. The swell of a wave in the ocean, and the subsequent dip that follows, is simply a vibration or oscillation of the water at the ocean's surface.

Electromagnetic waves are similar, but it consists of two waves oscillating perpendicular to each other. One is the oscillating **electric field** and the other one is an oscillating **magnetic field**. The direction of propagation of the wave is perpendicular to the electric field and the magnetic field (see figure 1.1). They are **transverse waves** and do not require a material medium for propagation. They travel at the speed of light which is 3 x 10^8 m/s.

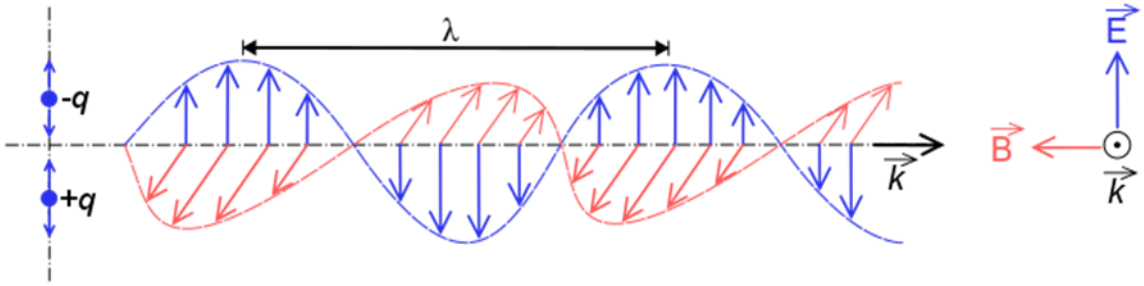


Fig 1.1 electromagnetic radiation

Basic Properties of Electromagnetic Waves

1. Electromagnetic waves are **transverse** in nature.
2. They propagate by varying **electric fields and magnetic fields**, such that these two fields are **at right angles** to each other and at **a right angle** with the direction of propagation of the wave.

3. Electromagnetic waves travel with a constant velocity in vacuum. The speed of the waves is **3 x 10^8 m/s.**
4. Electromagnetic waves are non-mechanical waves. They **do not** require any **material** medium for propagation.
5. Electromagnetic waves have Amplitude (A), wavelength(λ), frequency(ν)and period (T)

Amplitude, wavelength, frequency and Period

As you might already know, a wave has a trough (lowest point) and a crest (highest point). **The vertical distance between the top of a crest to the equilibrium position, or the position of the medium at rest known as amplitude which is the brightness, or intensity, of the wave**.

The horizontal distance between two repeated troughs or crests is known as the wavelength of the wave. As shown in figure 1.2

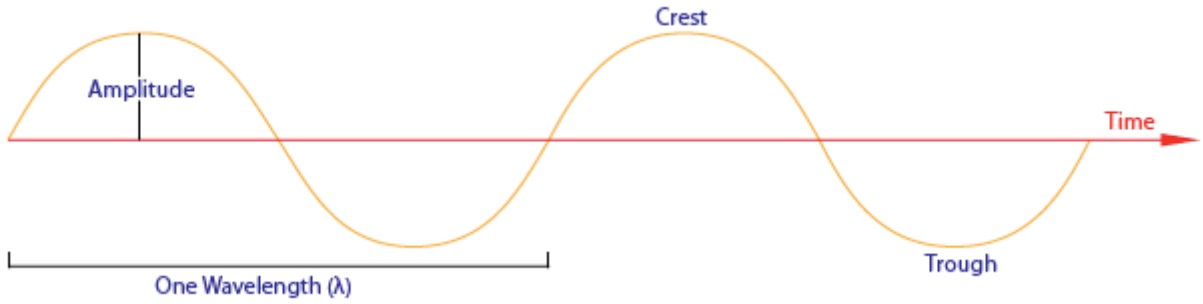


Fig 1.2

The quantity known as the wave's **frequency refers to the number of full wavelengths that pass by a given point in space every second**; the SI unit for frequency is **Hertz**, which is equivalent to "per seconds". As you might imagine, wavelength and frequency are inversely proportional: that is, the shorter the wavelength, the higher the frequency, and vice versa. This relationship is given by the following equation:

$c = \lambda \nu$ $\nu = C / \lambda$

where (the Greek lambda λ) is the wavelength (in meters) and (the Greek nu ν) is the frequency (in Hertz,). Their product is the constant, the speed of light (c) is given by **$c=3*10^8$ [m/sec]**

This relationship reflects an important fact: all electromagnetic radiation, regardless of wavelength or frequency, travels at the speed of light.

The last quantity we will consider is the period of a wave. A wave's **period is the time it takes for one wavelength to pass by a given point in space**. Mathematically, the period (T) is simply the reciprocal of the wave's frequency (ν):

$$T = 1/\nu$$

The units of period are second (s).

Now that we understand some basic properties of waves

The electromagnetic spectrum

Electromagnetic waves can be classified and arranged according to their various **wavelengths/frequencies**; this classification is known as the electromagnetic spectrum. Figure 1.3 show the spectrum, which consists of all the types of electromagnetic radiation that exist in our universe.

As we can see, the **visible spectrum**—that is, light that we can see with our eyes makes up only a small fraction of the different types of radiation that exist. To the left of the visible spectrum, we find the types of energy that are lower in frequency (and thus longer in wavelength) than visible light. These types of energy **include infrared (IR)** rays (heat waves given off by thermal bodies), **microwaves**, and **radio waves**. These types of radiation surround us constantly, and are not harmful, because their frequencies are so low. As we know, "the photon," with lower frequency are lower in energy, and thus are not dangerous to our health.

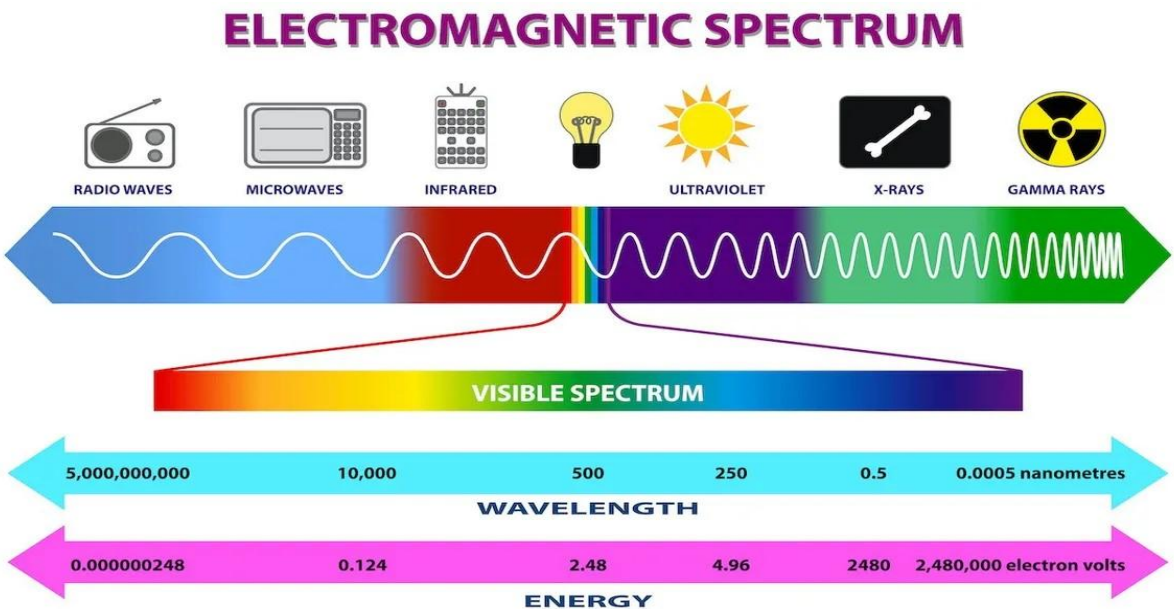


Fig 1.3 electromagnetic spectrum

To the left of the visible spectrum, we have **ultraviolet (UV) rays, X-rays, and gamma rays**. These types of radiation are harmful to living organisms, due to their extremely high frequencies (and thus, high energies). It is for this reason that we wear sun screen lotion (to block the UV rays from the sun) and why an X-ray technician will place a lead shield over us, in order to prevent the X-rays from penetrating anything other than the area of our body being imaged. Gamma rays, being the highest in frequency and energy, are the most damaging. Luckily though, our atmosphere absorbs gamma rays from outer space, thereby protecting us from harm.

Next, we will talk about the relationship between a wave's frequency and its energy.

1.3 Bohr model of the atom.

Lasing action is a process that occurs in matter. Since matter is composed of atoms, we need to understand (a little) about the structure of the atom, and its energy states.

We shall start with the **semi-classical model**, as suggested in 1913 by **Niels Bohr**, and called: **The Bohr model of the atom**.

According to this model, every atom is composed of a very **massive nucleus** with a **positive electric charge (=Ze)**, electrons are moving in specific paths around the nuclei.

Z = Number of protons in the nucleus, **e** = Elementary charge of the electrons:

Figure 2.1 illustrates a simple, but adequate, **picture of the atom, the Bohr model:** Every "**allowed orbit**" of the electron around the nucleus, is connected to a specific energy level.

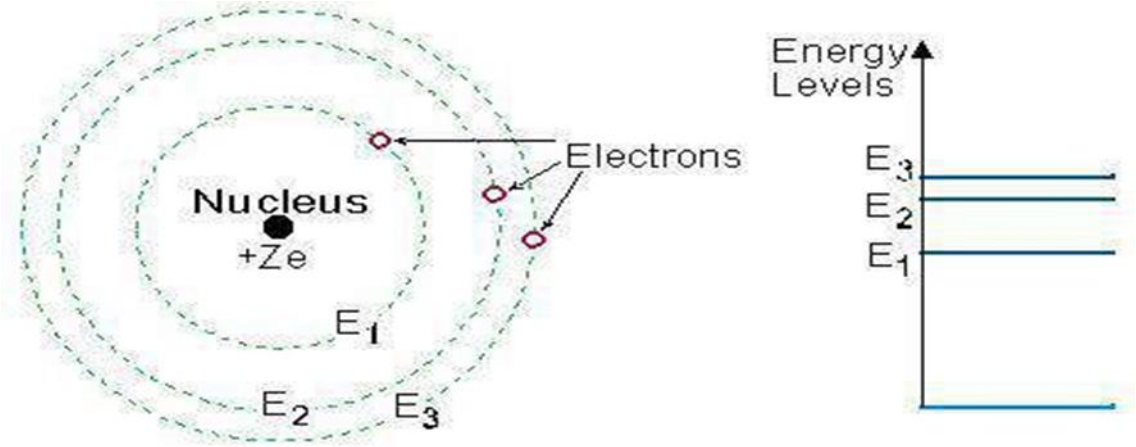


Fig 1.4: Bohr picture of the Atom

The energy level is higher as the distance of the "orbit" from the nucleus increases. Since for each atom there are only certain "allowed orbits", **only certain discrete energy levels exist**, and are named: **E_1, E_2, E_3**, etc..

Energy States (Levels) Every atom or molecule in nature has a specific structure for its energy levels. The lowest energy level is called the **ground state**.

As long as no energy is added to the atom, the electron will remain in the ground state

When the atom receives energy (electrical energy, optical energy, or any form of energy), this energy is transferred to the electron, and raises it to a higher energy level (in our model further away from the nucleus). The atom is then considered to be in an **excited state**. The electron can stay only at the specific energy states (levels) which are unique for each specific atom.

The **electron cannot be in between these "allowed energy states"**, but it can "jump" from one energy level to another, while receiving or emitting specific amounts of energy.

These specific amounts of energy are equal to the **difference between energy levels within the atom.**

Each amount of energy is called a "**Quantum**" of energy

(The name "**Quantum Theory**" comes from these discrete amounts of energy).

Energy transfer to and from the atom can be performed in two different ways:

1.Collisions with other atoms, and the transfer of kinetic energy as a result of the collision. This kinetic energy is transferred into internal energy of the atom.

2. Absorption and emission of electromagnetic radiation. Since we are now interested in the **lasing process,** we shall concentrate on the second mechanism of energy transfer to and from the atom

Photons and the energy diagrams

Electromagnetic radiation has, in addition to its wave nature, some aspects of "**particle like behavior**". In certain cases, the electromagnetic radiation behaves as an ensemble of discrete units of energy that have momentum. These discrete units (quanta) of electromagnetic radiation are called "**Photons**". The relation between the **amount of energy (E)** carried by the photon, and its **frequency (ν)**, is determined by the formula (first given by Einstein):

$$E = h\nu$$

This formula shows that **the frequency of the radiation (ν), uniquely determines the energy of each photon in this radiation.**

or with wavelength: $\mathbf{E = h * c/\lambda}$

where E is the energy of the photon absorbed or emitted (given in Joules), ν is frequency of the photon (given in Hertz), and ***h*** is Planck's constant= $6.625*10^{-34}$.

This formula shows that **the energy of each photon is inversely proportional to its wavelength**.

•The interactions between electromagnetic radiation and matter cause changes in the energy states of the electrons in matter.

•Electrons can be transferred from one energy level to another, while absorbing or emitting a certain amount of energy. This amount of energy is equal to the energy difference between these two energy levels (E_2-E_1).

For the sake of ease in understanding consider two energy states E_1 and E_2, of an atom. E_1 the lower energy state while E_2 is the excited state. As the constituent are identical. the energy states E_1 and E_2 Will be common to all atoms in the medium. Let a monochromatic radiation of frequency ν be incident on the medium. The radiation may be viewed as a stream of photons, each photon carrying an energy $h\nu$, The photon is given by energy of the emitted is given by

$\Delta E = E_2 - E_1 = h\nu = hc/\lambda$

There are three distinct processes through which interaction of radiation and matter can take place. They are: (1) Absorption (2) Spontaneous emission and (3) Stimulated emission.

Example/ Calculate **the energy difference in eV** between the two energy levels of the Ne atoms of a He - Ne gas laser, the transitions between which results in the emission of a light of wavelength 632.8 nm. Also, calculate the number of photons emitted/second, if the optical power output is 2 mW.

Data: Wavelength, $\lambda = 632.8$ nm, P = 2 mW.

To find: Energy difference, ΔE and number of photons **N**

Solution:

1.3 Characteristics of Laser light

light has four unique characteristics that distinguish it from ordinary light: these are;

1. Coherence
2. Directionality
3. Monochromatic
4. High intensity

1. Coherence

Coherence is a relationship between the amplitude and phase of a light wave at any point to the other point in space or time.

Light that is emitted from **normal light sources** is in the form of short random wave packets which have low coherence. But Lasers Coherence length (L_{cohe}) is usually a few kilometers. Quite often the coherence length (L_{cohe}) is in the range of millimeters to centimeters in ordinary monochromatic light.

$$L_{cohe} = \lambda_o^2/(n\Delta\lambda)$$

λ_0:central wave length the mean or peak wave length

$\Delta\lambda$: spectral line width

*n:*refractive index

There are two types of coherence: (i) Temporal Coherence and (ii) Spatial Coherence.

(i) Temporal Coherence: In practice, no light source is ideally monochromatic. The wavelength of the light radiations emitted by a monochromatic source have certain variation in wavelength. If the variation in wavelength is less, the temporal coherence of the light is said to be high. Since laser light is highly monochromatic, its temporal coherence is very high.

It is directly related to the spectral width (monochromaticity) of the light source. Smaller the change in wavelength ($\Delta\lambda$), more temporal coherent the light.

(ii) Spatial Coherence: At any instant of time, if the phase difference between any two points separated spatially in a wave separation in any of the waves emitted by the same light source, then light has spatially coherent.

When the phase difference between any two points in the wave front does not change with time, the beam is said to be spatially

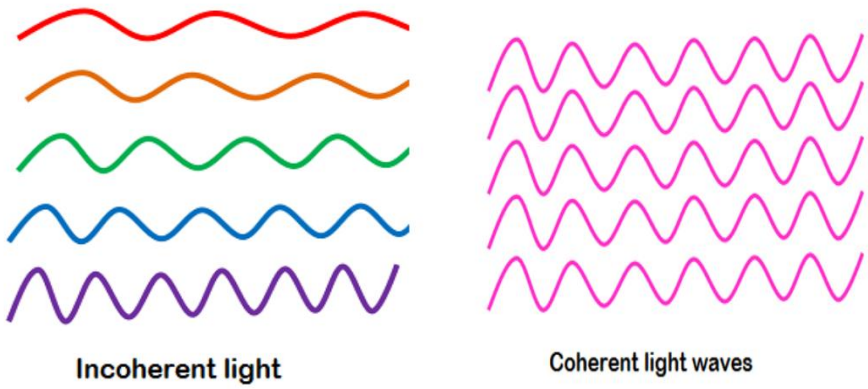


Fig 1.5 coherent and Incoherent light

Example/ Imagine that we chop a continuous laser beam (assumed to be perfectly monochromatic λ_0= 623.8 nm) into 0.05 pulses using some sort of shutter. Compute the resultant line width $\Delta\lambda$ and coherence length.

Data: Δt = 0.05 \times ns, λ_0= 623.8 nm

To find: Resultant line width and coherence length

Solution:

2. Directionality

In conventional light sources (lamp, sodium lamp), photons will travel in random direction. Therefore, these **light sources emit light in all directions.**

On the other hand, in laser, all photons will travel in same direction. Therefore, laser emits light only in one direction. This is called directionality of laser light. The width of a laser beam is extremely narrow. Hence, a **laser beam can travel to long distances without spreading.**

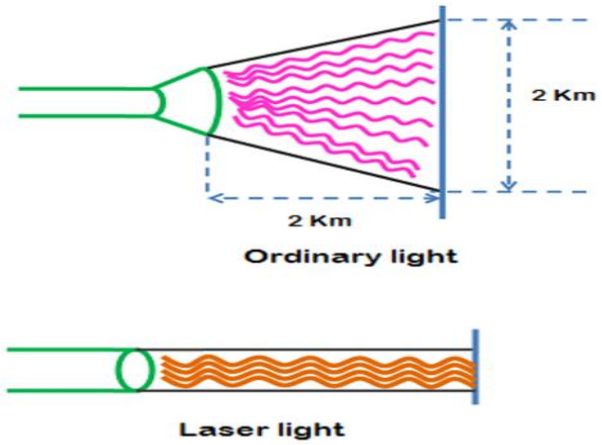


Fig 1.6 Directionality of laser

If an ordinary light travels 2 km, it spreads to about 2 km in diameter. On the other hand, if a laser light travels 2 km, it spreads to a diameter less than 2 cm.

3. Monochromatic

Monochromatic light means a light containing a single color or wavelength. The photons emitted from ordinary light sources have different energies, frequencies, wavelengths, or colors. Hence, the light waves of ordinary light sources have many wavelengths or colors. **Therefore, ordinary light is a mixture of waves having different frequencies or wavelengths.**

On the other hand, in laser, all the emitted photons have the same energy, frequency, or wavelength. Hence, **the light waves of laser have single wavelength or color.** Therefore, laser light covers a very narrow range of frequencies or wavelengths.

4. High Intensity

You know that the intensity of a wave is the energy per unit time flowing through a unit normal area. In an ordinary light source, the light spreads out uniformly in all directions. If you look at a 100-Watt lamp filament from a distance of 30 cm, the power entering your eye is less than 1/1000 of a watt.

In laser, the light spreads in small region of space and in a small wavelength range. Hence, **laser light has greater intensity when compared to the ordinary light.**

If you look directly along the beam from a laser (caution: don't do it), then all the power in the laser would enter your eye. Thus, even 1 Watt laser would appear many thousand times more intense than 100 Watt ordinary lamp.

Thus, these four properties of laser beam enable us to cut a huge block of steel by melting.

Intensity I of light (often called irradiance) is given by

Intensity (I) = Power (P) /Area (A) , usually in units of W

The power of a laser beam is the rate at which the beam delivers optical energy. It is the ratio of energy to time.

Power (P) = Energy /Time, usually in units of J/s or W

Since intensity (irradiance) I = P/A and the area,

if circular, is $A = \pi r^2$, $r=d/2$

$A=\pi(d/2)^2 =\pi d^2/4$

$I=4P/\pi d$

Example/Find the intensity of a laser beam of 20 mW power and having a diameter of 1.3 mm. Assume the intensity to be uniform across the beam.

Data: Power, P = 20mW, d = 1.3 mm

To find: Intensity of laser beam I

Solution:

Components of Lasers

1. Active Medium:

   It is the material in which the laser action takes place. The active medium may be solid crystals such as ruby or Nd:YAG, liquid dyes, gases like CO2 or Helium / Neon, or semiconductors such as GaAs. This medium decides the wavelength of laser radiation. Active mediums contain atoms which can produce more stimulated emission than spontaneous emission and cause amplification they are called “Active Centers”.

2. **Pumping Energy Source (Excitation Mechanism):**

   Energy Source (Excitation mechanisms) pumps the active centers from ground state to excited state to achieve population inversion. The pumping by energy source can be optical, electrical or chemical depending on the active medium.

3. **Resonance Cavity:**

   Resonance cavity consists of active medium enclosed between two mirrors one is highly reflective mirror (100% reflective) and the other is partially transmissive mirror (99% reflective

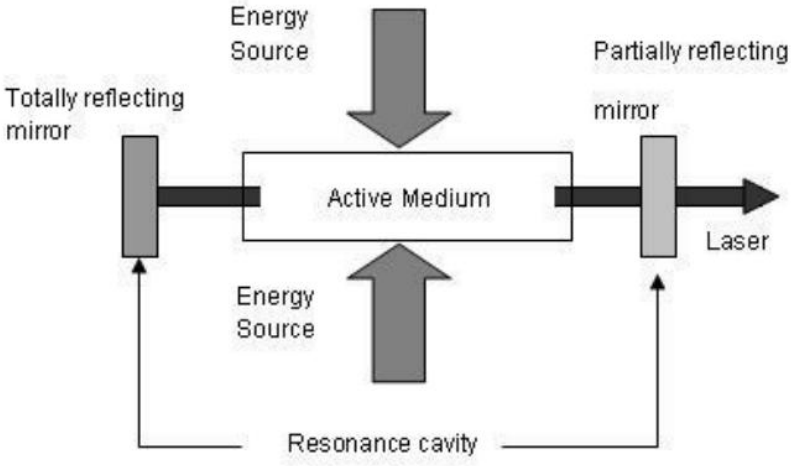


Fig 1.7 component of laser